

Raw SERS data



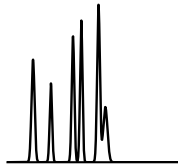
SERS map

one spectrum
per pixel



NNLS gate

background
dropped

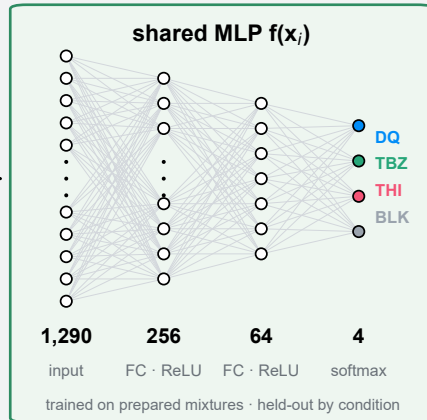


pixel spectra x_i

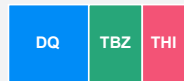
1,290 pts · 500–2500 cm^{-1}

Compositional unmixing

output



map composition



gated pixels · composition

$\bar{p}_i = \text{pixel mean} \cdot \text{BLK removed}$

absent component → reported 0

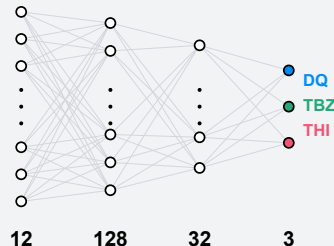
(absence gate, $P < 0.2$)

Concentration

output

band leq · composition · intensity (3 ea + p10/50/90)

concentration net $g(\text{map})$



map features FC · ReLU FC · ReLU μM

validated 3–24 μM · outside → pixel-library k-NN (measured-only) · else no answer

declared total (optional)

$\hat{C}_i = \bar{p}_i \cdot C_{\text{total}}$ — overlay when known